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Submission ID**trn:oid::1:977284007****Submission Date****Jul 1, 2026, 6:40 AM GMT+5****Download Date****Jul 1, 2026, 6:44 AM GMT+5****File Name****final.docx****File Size****1.4 MB****5 Pages****2,475 Words****16,087 Characters**

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From Raw Data to Business Intelligence: A Data Engineering Framework for Online Retail Analytic

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Abstract—The rapid growth in the amount of retail transaction data has resulted in a high demand for an efficient data engineering approach to analyze the data for business intelligence. However, there are issues of inconsistencies, duplicates and missing data which make the transactional data difficult to use for analysis. This paper presents how an end-to-end data engineering pipeline can be built using Online Retail Transaction data. The proposed data engineering approach makes use of the Extract, Transform and Load (ETL) approach, wherein Python will be used for pre-processing the data, Pentaho Data Integration (PDI) for transformation and automation of the workflow, MySQL as data warehouse and Microsoft PowerBI for interactive data visualization. The processed data is then organized into fact and dimension tables in a star schema to enable efficient analytical queries and reporting. ETL is an efficient process for converting raw transactional data into a tidy, organized and reliable data set that can be used for sales analysis, customer behavior analysis and product performance analysis. The resulting dashboard shows that the proposed pipeline is effective in preparing retail transaction data for business intelligence applications and providing meaningful insights to support informed business decisions.

Keywords—Extract, Transform, and Load (ETL), Pentaho Data Integration, Data Warehouse, Star Schema, Business Intelligence, Power BI

I. INTRODUCTION

The massive growth of e-commerce has led to the exponential growth of retail transaction data generated on daily basis. However, raw transactional data are often characterized by duplicate records, missing values and inconsistent information, which are not suitable for analysis. Thus, a data engineering pipeline is required to transform raw data into structured and trustworthy datasets that enable business intelligence and data-driven decision making [1].

The dataset used in this project is the Online Retail Transaction dataset which contains transaction data of a UK-

based online retail company. The goal of this project is to create an end-to-end ETL (Extract, Transform, Load) pipeline to prepare the dataset for analysis. Python for preprocessing, Pentaho Data Integration for ETL, MySQL for storage, and Microsoft Power BI for visualization, transforming raw transactional data into a clean and structured data warehouse. This allows for more efficient analysis of sales trends, customer behaviour and product performance, and offers robust insights for business decision-making [2].

II. DATASET DESCRIPTION

The database for the project is Online Retail Transaction database has been taken from Kaggle. It consists of 541,909 transaction data from an online retail company based in the UK from Dec 2010 - Dec 2011. The dataset contains several attributes including InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country, providing an insight into sales transactions and customer purchases [3]. The dataset is useful for analyzing the retail sales, but it suffers from data quality issues, such as duplicate records, missing values and invalid transaction records. To solve these, an ETL pipeline was designed for data cleaning, transforming and structuring the data into a data warehouse. The resulting dataset serves as a starting point for analyzing sales trends, behavior of the customers and performance of products.

III. METHODOLOGY

The proposed work involved designing and development of a project using an Extract, Transform and Load (ETL) approach. The process involves six major steps: data ingestion, data storage, data transformation, data modeling, data quality validation, and automation of ETL process. The complete process is aimed at transforming the raw transaction data into

structured data warehouse for business intelligence and analytical reporting [4].

A. Data Ingestion

The dataset is read into Python for data pre-processing operations like removal of duplicate entries and handling of missing data entries. Following data pre-processing, the cleaned dataset is loaded into MySQL staging table, which is an input for ETL process in PDI.

B. Data Storage

MySQL is used as the relational database management system to store both the raw data as well as the data warehousing. The staging table holds the cleaned up transaction data while the data warehouse holds the fact and dimensional tables. This architecture ensures an organized manner of storing data and processing queries.

C. Data Transformation

The data transformation phase is to transform the preprocessed data into the structure that fulfills the requirements of the data warehouse. The complete ETL process is executed with Pentaho Data Integration (PDI) through four transformations: `dim_product`, `dim_customer`, `dim_time`, and `fact_sales`. The process includes attributes selection, derived attributes creation, data validation, sales metrics calculation, and data mapping into a dimensional structure before loading into the database.

1. Product Dimension Transformation

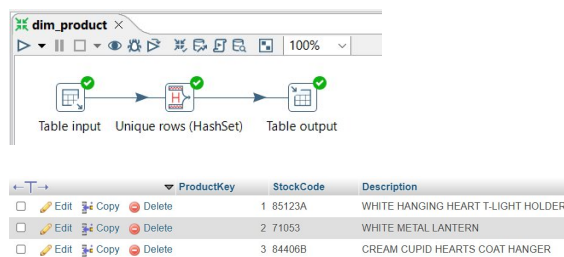


Fig. 1. Product dimension transformation workflow and output table.

The `dim_product` transformation will create a product dimension table that will be used as reference in the fact table. The process starts with the Table Input in Pentaho Data Integration. It reads data from the `onlineretail_clean` table with the SQL query. This query is only fetching the attributes `StockCode` and `Description` required to build the product dimension. Then, Unique Rows (HashSet) process is used to validate the data and make sure there are no duplicate product records before loading into the database.

Then the data is validated and loaded into the table `dim_product` using a Table Output. At this point, a `ProductKey` (a surrogate key) is created via a SQL query setup in the Table Output, which guarantees that each product has a unique identifier to be referenced in the fact table.

2. Customer Dimension Transformation

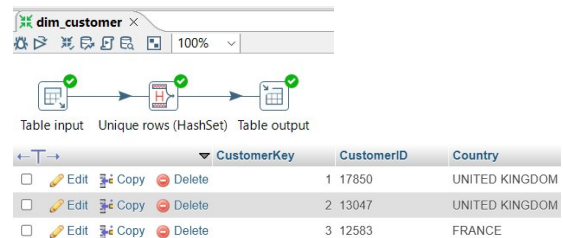


Fig. 2. Customer dimension transformation workflow and output table.

The `dim_customer` transformation aims to build a customer dimension table as a reference for the data analysis process. The process begins with the Table Input in Pentaho, which reads data from the `onlineretail_clean` table using an SQL query that selects the `CustomerID` and `Country` attributes as required by the customer dimension. After that, Unique Rows (HashSet) is used as a validation process to ensure there are no duplicate customer records.

The validated data is then loaded into the `dim_customer` table using a Table Output. The `CustomerKey` is generated as a surrogate key via an SQL query configured in the Table Output, ensuring that each customer has a unique identifier to be linked to the fact table.

3. Time Dimension Transformation

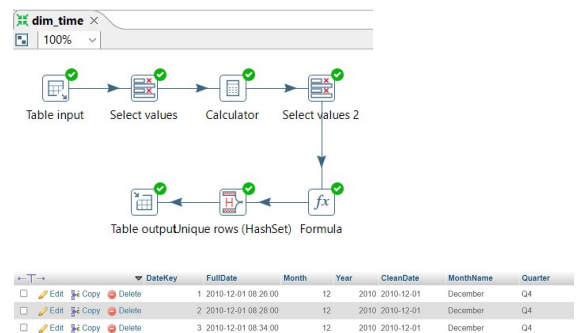


Fig. 3. Time dimension transformation workflow and output table.

It begins with the Input Table in Pentaho that reads the `InvoiceDate` attribute from the `onlineretail_clean` table with a SQL query. The Select Values then renames the attribute to `FullDate` and changes the data type from Timestamp to Date. The Calculator then creates the `CleanDate`, `Month` and `Year` attributes based on the values in the transaction date. Select Values is used to reformat the `CleanDate` column to year-month-day format. Then a Formula is used to generate the `MonthName` and `Quarter` attributes, so that the time information can be used for analysis based on various periods.

The validation with 'Unique Rows' (HashSet) is performed before the data loading in order to verify that each date is only present once in the dimension table. The transformed data is loaded into the `dim_time` table using Table Output where `DateKey` is created as a surrogate key using a SQL query configured in Table Output.

4. Fact Sales Transformation

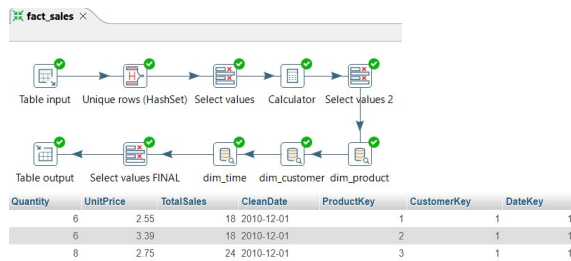


Fig. 4. Fact Sales transformation workflow and output table.

The fact_sales transformation is an integration step which creates a fact table to act as a central repository of transaction data. It starts with Table Input. Then Unique Rows (HashSet) is used to validate before joining the tables. Select Values is used to choose the attributes you want to select and change data types. Then, the TotalSales value is calculated by Calculator based on the product of Quantity and UnitPrice, and the second Select Values is used to adjust the CleanDate attribute as a reference for the lookup process in the time dimension.

The integration with the dimension tables is done through three Database Lookups: dim_product, dim_customer, and dim_time to retrieve the ProductKey, CustomerKey, and DateKey based on the values of the corresponding natural keys. Finally, after all surrogate keys have been obtained, Select Values is used again for the final selection process before data is loaded into fact_sales table using Table Output.

D. Data Modeling

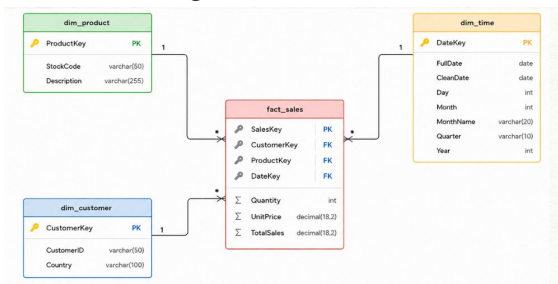


Fig. 5. Star Schema of the proposed retail data warehouse.

In this project, the data warehouse model is based on the star schema approach, consisting of one fact table (fact_sales) and three dimension tables (dim_product, dim_customer, and dim_time). This approach was chosen for its simple structure, less data redundancy and efficient support for multidimensional analysis.

Quantity, UnitPrice and TotalSales are the measures which are numerical values used in the fact_sales table which contains the transaction data. This table also contains foreign keys ProductKey, CustomerKey and DateKey to link each transaction to the dimension tables. Meanwhile, dim_product stores information about the product, dim_customer stores information about the customer, and dim_time stores date, month, year, month name, and quarter. This separation of data into fact tables and dimension tables allows the data to be more efficiently aggregated, filtered and visualized in the analysis phase.

E. Data Quality

To ensure data reliability, several validation procedures are applied throughout the ETL process. These include duplicate removal, missing value handling, data type validation, and verification of transaction values. Records with invalid quantities or unit prices are excluded to maintain the accuracy and consistency of the analytical dataset [5].

F. ETL Automation

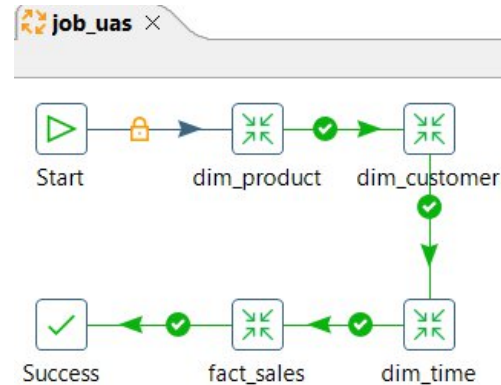


Fig. 6. Pentaho Job for ETL Process Automation.

The whole transformations are realized in an integrated way of a Pentaho Job. The job controls the execution order of the transformations from dim_product, dim_customer, dim_time to fact_sales. This order is to make certain that all dimension tables are created first so that the database lookup process in the fact_sales transformation can always get the required surrogate keys.

IV. RESULTS AND DISCUSSION

The proposed ETL pipeline was successfully implemented using Python, Pentaho Data Integration, MySQL, and Microsoft Power BI. This section presents the implementation results and discusses how the processed data were organized and build into a structured data warehouse for business intelligence analysis.

A. Data Warehouse Implementation

Table	Action
☐ dim_customer	Browse Structure Search Insert Empty
☐ dim_product	Browse Structure Search Insert Empty
☐ dim_time	Browse Structure Search Insert Empty
☐ fact_sales	Browse Structure Search Insert Empty
☐ onlinereetail_clean	Browse Structure Search Insert Empty
5 table(s)	Sum

Fig. 7. Database tables created after the ETL Process.

The ETL process successfully organized a structured data warehouse in MySQL. As shown in Fig. 7, the database consists of one staging table (onlinereetail_clean), three dimension tables (dim_product, dim_customer, and dim_time), and one fact table (fact_sales). The table stores the cleaned transactional data before it is transformed into the dimensional model. The dimension tables contain descriptive information, while the fact table stores transactional measures used for analytical reporting.

B. Business Intelligence Dashboard

The clean data stored in MySQL was combined with Power BI to make an interactive and informative dashboard that will provide company stakeholders with a clear information of the business performance that will help in decision making later. The complete dashboard visualization is shown in Fig.8.

Fig. 8 shows performance indicators (KPI) like total revenue which is 8.72M, come from a catalog of 3,665 products and 4,338 customers. The United Kingdom take 92.4% of all transactions, showing where the company's market is concentrated. Fig. 8 tracks sales trends and customer behavior over time which the "total customers by year and month" view shows that the regular customer(old) is making more transaction than newcomers, pointing to strong customer loyalty. Sales activity spikes from October-November because of year holiday shopping for Christmas and New Year. In the "Total Revenue by Year and Month" view there is a rising revenue each month indicating a healthy revenue. Customer many buy on work day which Thursday become its peak day. "Total Transaction and Total Revenue by CustomerID" in Fig.8 shows that the relationship between transaction frequency and total spending, showing high-value customers who contribute disproportionately through purchases frequency.

which columns represent total revenue and the line represents quantity sold. This makes it easy to spot the highest profiting products, "PAPER CRAFT, LITTLE BIRDIE," and products with high sales volume which is informing its need for inventory planning and marketing strategy. Fig. 9 also include interactive slicer to let stakeholders search specific items for detailed analysis. On below, 2 horizontal bar charts show item performance: the green showing the top revenue products, the red showing the lowest performing products with almost 0 sales. This helps stakeholders decide which products to prioritize for restocking or discontinue.

C. Analytical Insights

The business make a total revenue of £8.72M from 4,338 customers and catalog of 3,665 products. The dominated country is the UK (92.84% of transactions), with revenue rising sharply from October-November 2011 because of the holiday season, and sales activity many done on weekdays, peaking on Thursday while dropping off on weekends. Old customers outnumber newcomers every month, pointing to a loyal customer. In Fig. 8 shows most customers is near zero which mean they made infrequent and low-value transactions. Some customers (outliers) did many transaction / at high volume, contributing disproportionately to total revenue. On the product side, "PAPER CRAFT, LITTLE BIRDIE" made the most profit, while "LETTER 'B' BLING", "MINI JIGSAW", and many more make near-zero revenue.

V. CONCLUSION & REFLECTION

In this project, a complete data engineering pipeline has been designed that can turn the dirty online retail transaction data into clean data and make it available for analysis. The ETL process has been used in this project that involves the use of python and Pentaho Data Integration to perform operations related to data cleaning such as handling duplicated records, invalid data, and missing value. After the cleaning, data modeling has been done by creating Star Schema model and storing it in the MySQL data warehouse for doing analytical queries. With the integration of data warehousing and Microsoft Power BI, the interactive dashboard has been created to provide insights such as UK market leader, high customer loyalty, seasonality effect in buying behavior, and performance of the products. This is how the data pipeline turns the raw operational data into reliable source of data analytics. In this project, the importance of data quality in Business Intelligence has been stressed as otherwise the dirty data will result in wrong visualization. The main challenge in the project is the creation of Star Schema as this requires understanding of the business requirements to create facts table and dimensions.

For Fig. 9 focus on product analysis which has a combo chart (line and clustered column) shows the top 20 products,

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